

FYP Proposal

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1. Title

A Clinical Decision Support System for Prescription Risk Screening using Graph-LLM Integration

2. Introduction / Problem Statement

Patients with heart and circulation problems typically have complex medical histories. They could be diagnosed with more than one condition at once and be receiving a host of medications. They could also have different test results from the last time they went. The doctor must examine the entire situation before deciding on the medication, rather than only reading the name of the medication. Presently, current hospital systems for prescription safety only undertake basic name lookups. They compare the proposed drug to a list of drug combinations that are bad. The checking process just stops right there. This approach is only suitable for simple, well-known interactions of the drugs. Fails to take into account other significant detail within complex patient scenarios. Physicians make notes in normal text sections. A physician may record the patient's recent eGFR level of less than 30. They may also enter a new discovery that's been made during a recent scan. These are typed notes which the normal database cannot read. But the other is indirect risk. A drug might seem safe by itself. It could actually be dangerous based on the patient's specific health conditions and other drugs and lab results. Fixed tables cannot connect all these different pieces of information. They are simply not designed to link things together in that way.

LLMs can read normal clinical text. They might seem like a good way to solve the reading problem. The main issue here is hallucination (Pal et al., 2023). These models can make up statements. The statements sound very professional but are actually

wrong sometimes. The models cannot tell the difference between real facts and made-up details. Making up a fake drug interaction is a huge problem for patient safety. A system making up false warnings creates a brand new problem instead of fixing the original issue.

The new system design avoids this issue completely. Each part of the system does a specific job to avoid weaknesses. A small open-source LLM only reads the input text. It looks at the checked boxes and typed notes from the doctor. The LLM then changes this text into Python query parameters. The LLM does not make any medical decisions during this step. These parameters are used to run Cypher queries. They search a Neo4j knowledge graph containing drug safety data from DrugBank and fake patient records from Synthea. The graph examines the relationships between drugs and diseases and laboratory findings. It searches for direct risks and also for indirect risks that are not found by simple searches. The graph is the only thing that can provide real data. It can't make things that are not connected connect. These results are used in a Python script that is calling on the MERIS formula (Saedder et al., 2015). It determines a risk number and displays the precise connections which led to the risk. The LLM then changes this data into a simple warning message for the doctor. The staff of the clinic simply communicate with the system in normal language. All details in the last warning refer to a true connection in the graph data.

3. Aims and Objectives

3.1 Aim

This project is aimed at development and validation of clinical support system. The system is based on a small language model that reads the text and connects it to a Neo4j knowledge graph. This configuration detects combinations of dangerous drugs and computes a MERIS risk score. It gathers data from doctor input and synthetic

patient data generated by Synthea.

3.2 Objectives

- Develop a Text Reading module using an open source LLM less than 10 billion parameters. In this module, we will transform common medical notes into Python Cypher queries without additional model training using specific prompts.
- Construct a NeO knowledge graph with medical information. All Drug Interaction and Disease Warning data is from DrugBank and SIDER 4.1, respectively. The system will also populate the graph with patient histories created by Synthea to enable the system to traverse complex medical relationships.
- Write a Python script to compute the risk score (MERIS) from the result of the graph search. This script uses the same scoring rules and cut-offs as published by Saedder et al. (2016).
- Run tests to evaluate the system and compare with a normal LLM. The tests will confirm accuracy of identifying drug risks and how many times it will fabricate information in specific test cases that can be answered.

4. Literature Review

This literature review looks at the issues that arise with the need to improve prescription checking tools in clinics. Discusses pros and cons of the use of LLMs to read medical text. It also provides tech explanation why LLM and KG are combined to make medical software.

4.1 The Clinical Context of Prescription Errors

Many adults in the hospital are injured by their medications. This is a very common clinical problem. Lam et al. (2026) observed that many of these adverse drug incidents can be avoided by hospitals. The most dangerous group are those who are using a large number of medications. Heart care is particularly hazardous. Heart patients typically are prescribed combined medications, such as antihypertensive, anticoagulant, and lipid-lowering drugs. Modern hospital software works to prevent this damage. The systems alert physicians to dangerous drug interactions when filling a prescription. The majority of these programs rely on databases containing a list of the most common drug names and warning abbreviations to check the name of a drug that is proposed. Simple or known drug matches work well with this method. These database records lack two important categories of dangers, and are not able to read common typed notes from doctors. These text boxes are used by doctors to record recent lab tests, scans or previous illnesses. The databases are also unable to identify indirect risks. It may appear that a drug is entirely harmless. The same medication may be harmful if combined with a certain disease, kidney problems and other common medications. Physicians end up with too many bad indications. It's a very common problem that is referred to as alert fatigue. Physicians begin to tune out the numerous poor quality notifications. The real dangerous combinations of drugs then seep through unnoticed.

4.2 LLM Capabilities and Limitations in Medical Text

LLMs have been applied to medical text processing. LLMs are currently being applied to medical text processing. These models can perform functions that fixed databases can't. Normal human language was easily understood by them. Abumelha et al. (2025) conducted a study on a two-step LLM framework. This system extracts medical information from doctors' notes. It works much more accurately than a one-step approach. The difficulty arises with the models when they are asked to think logically

in the main problem. There has been testing of LLMs against BioBERT models by the Gopalakrishnan et al. (2024). They wished to evaluate the performance of the models in detecting medical causes/effects. The LLMs generate extremely fluent sentences. In still finding connections that require a strict logic, they still make more mistakes. Hallucination is the worst difficulty in taking medical decisions. Pal et al. (2023) developed a test, called Med-HALT test, to assess this particular condition. They tried a lot of different LLMs! Many of the models were able to produce statements that seemed very professional. Those statements were entirely incorrect as far as the facts were concerned. Prescription checking systems can't create bogus drug warnings. This is no minor error. A high risk to patient safety. This gives a hard constraint on the software design. The LLM is only allowed to perform simple language tasks. It should only read text and extract words, and format the final message. A different and reliable system must provide all the actual medical facts.

4.3 Knowledge Graphs for Clinical Decision Support

Knowledge Graphs (KGs) can perform functions that are not possible with traditional databases. They demonstrate the relationship between drugs, illnesses, patient information and lab results. KGs create a graph (network) that we can search through. Normal SQL searches only match rows that have set rules. Neo4j can connect through several steps in a Cypher query to locate hidden associations. This jumping technique is just as suitable as the build up of a medical threat from numerous interacting components. Zhou et al. (2022) demonstrated this in the real world. They created a warning system for hypertension medications with the help of a knowledge graph. They demonstrated that the graph was able to deal with the treatment rules very well in comparison to a standard database. Lin et al. (2023) went even further and examined the overall risk of drugs. They used graph thinking to find hidden drug dangers where simple search methods completely miss these hidden dangers. This system also comprises of the medical history of the patient. Lyu et al. (2023)

constructed a special graph for kidney disease related to diabetes. They used points in their graph for past illnesses and lab test numbers as nodes in the network. They demonstrated that it is possible to feed a patient's medical circumstances directly into the graph. This situation can then be matched with the drug data and the system can search for it. This graph can be created using the basic data and warnings for a drug reaction from DrugBank (Wishart et al., 2018). This is a database that is checked for drug-drug interaction and drug targets. Data on the adverse effects of drugs on specific diseases is available in SIDER 4.1. The graph setup allows the system to traverse all of this information and a plain old flat table doesn't.

4.4 Graph-LLM Integration (GraphRAG)

Retrieval-Augmented Generation (RAG) links LLMs to outside data sources. This gives the model access to real checked facts before it writes an answer. Lewis et al. (2021) created the RAG system to handle jobs that need a lot of facts. A search tool goes and gets helpful documents. The LLM then thinks about those documents instead of just guessing from its own memory. This method stops the model from making up fake answers for factual questions. Normal RAG still has a hard time answering questions that need to join many scattered pieces of proof. Checking risks for patients on many drugs requires exactly this type of joining. GraphRAG fixes this problem. It takes away the document searcher and uses a knowledge graph instead. Edge et al. (2025) introduced this new way to handle complicated questions that involve many different things in organized data. The system doesn't only extract text blocks, it actually walks along the paths in the graph. Then it provides a small part of this checked graph to the LLM to give background information. The LLM then types an answer. All its claims are restricted to actual links in the graph. In medicine this is a good example to illustrate, the AUGMERIS software (Bertalan et al., 2025). This tool will automatically calculate the MERIS risk score using AI. The AUGMERIS is a proof-of-concept application that demonstrates the effectiveness of drug risk scoring

via AI. The issue is that it does not traverse the graph, but only reads information from one of the tables. It is also not capable of reading health practitioners' handwritten notes. It also can't read regular handwritten notes from doctors. Nygren et al. (2026) provide additional evidence of the effectiveness of this approach. They conducted a drug safety monitoring test. They employed a GraphRAG configuration with small LLMs running on top of the Neo4j graph store. These models included Qwen-2.5-7B and Llama-3.1-8B. They achieved a 99.95-99.96% accuracy on detecting drug side effects. A normal LLM without a search tool only achieved about 63%. This definitely backs the primary design decision of this undertaking. The system should only allow the LLM to read text and write the final output message, and also must allow the graph to do the task of finding the actual medical facts.

4.5 Prescription Risk Quantification: The MERIS Framework

There needs to be a method of quantifying the threat identified by the graph. Saedder et al. (2016) created and checked the Medication Risk Score (MERIS). There are four components to MERIS that it uses to guess the likelihood of a patient being drug impaired. These parts include kidney health, number of drugs per day, risk of the respective drugs, and drug interaction. Høj et al. (2021) have investigated MERIS in general clinic patients with numerous medications. They determined that with a score of 15, it is a good indicator of danger. Their test showed an accuracy number of $AUC = 0.70$. The AUGMERIS software (Bertalan et al., 2025) demonstrates that math can be done automatically using a computer program. Our new system takes this idea a step further. It maps the drug mixing danger part to the search results of Neo4j. The system takes into account the depth and severity of the relationships present in the graph. This will generate a score, not just a flat list, based on the patient.

4.6 Summary of Research Gap

The research shows a clear split in the technology. LLM tools are great at reading normal language. They simply cannot be trusted to make real medical choices. Knowledge graph systems think very accurately about drug facts. They cannot read normal typed text. Current scoring tools like AUGMERIS mix math programs with AI. They do not travel through graphs and they cannot read messy doctor notes. No current system fixes all three of these problems at the same time. The new system design fixes this by giving each part a job it does best. The LLM only reads the start text and writes the final message. Neo4j handles all the deep factual searching. A Python script does the MERIS math on the graph results. Every final choice the system makes links straight back to a real database connection or a set math number. Table 1 provides an overview of three separate capability gaps that cannot be addressed by existing approaches. While standalone LLMs excel in parsing unstructured clinical text, they fail to reliably do deterministic multi-hop reasoning and tend to hallucinate in medical inference tasks (Abumelha et al., 2025; Gopalakrishnan et al., 2024; Pal et al., 2023). Traditional knowledge graphs have been able to handle a structured multi-hop traversal (Lin et al., 2023; Zhou et al., 2022), but lack any processing of free-text clinical notes as input. To date no scoring tool available can calculate the MERIS and can traverse the graph paths without being provided with previously structured clinical notes. The proposed system combines and assigns each capability to the component that is best suited for it: the LLM parses the text only, the Neo4j component handles the factual multi-hop reasoning, and a Python module applies the fixed MERIS formula on the results coming from the graph.

Table 1. Capability comparison of existing approaches against three clinical requirements addressed by the proposed system

	Unstructured Text Parsing	Multi-hop Reasoning	Automated MERIS Scoring
Standalone LLM	✓	×	×
Traditional KG	×	✓	×
Existing Scoring (e.g., AUGMERIS)	×	×	✓
Our Proposed System	✓	✓	✓

4.7 Literature Review Synthesis Matrix

A summary table maps out the main research papers for the technology areas talked about here. The full table is placed in Appendix A to make it easier to read.

5. Research Methodology & Ethics

5.1 Research Design and Theoretical Framework

This project uses an experimental approach to build the system. The main theory behind it is the Graph Retrieval-Augmented Generation (GraphRAG) framework. The independent variable is the system design. This compares the new graph and LLM combination against a normal LLM working alone. The dependent variables are the accuracy in finding drug risks, the correctness of the MERIS math, and how often the system makes up fake details. The graph rules strictly control all the medical facts. The LLM only handles reading and writing words.

5.2 System Architecture and Scope

The software interface only deals with the clinic staff. The system is not used by patients at all. This prototype remains totally isolated from actual patient information, to prevent ethic and practical problems. Rather, synthetic data from fake Synthea is used for all patient information, and DrugBank and SIDER 4.1 are used to provide the drug safety rules. The actual working of the system can be divided into three main layers. Staff enter a new prescription and any normal clinical notes during the parsing phase. This text is then read by a small open-source LLM which transforms it into a structured Python variables. The reasoning layer on top of Neo4j follows. Here, a Python script converts those variables into Cypher queries to enable the graph to search among drugs, patient history, and lab numbers. This particular step identifies both obvious drug risks and subtle indirect dangers which are not typically detected from simple searches. Finally, the scoring/output layer uses both Python and the LLM. The LLM translates the final numbers and graph paths from a Python script that was run on the search results, into a clear warning message for the doctor, using the MERIS formula (Saedder et al., 2016). The whole data process and how all these parts relate to each other is presented in Figure 1.

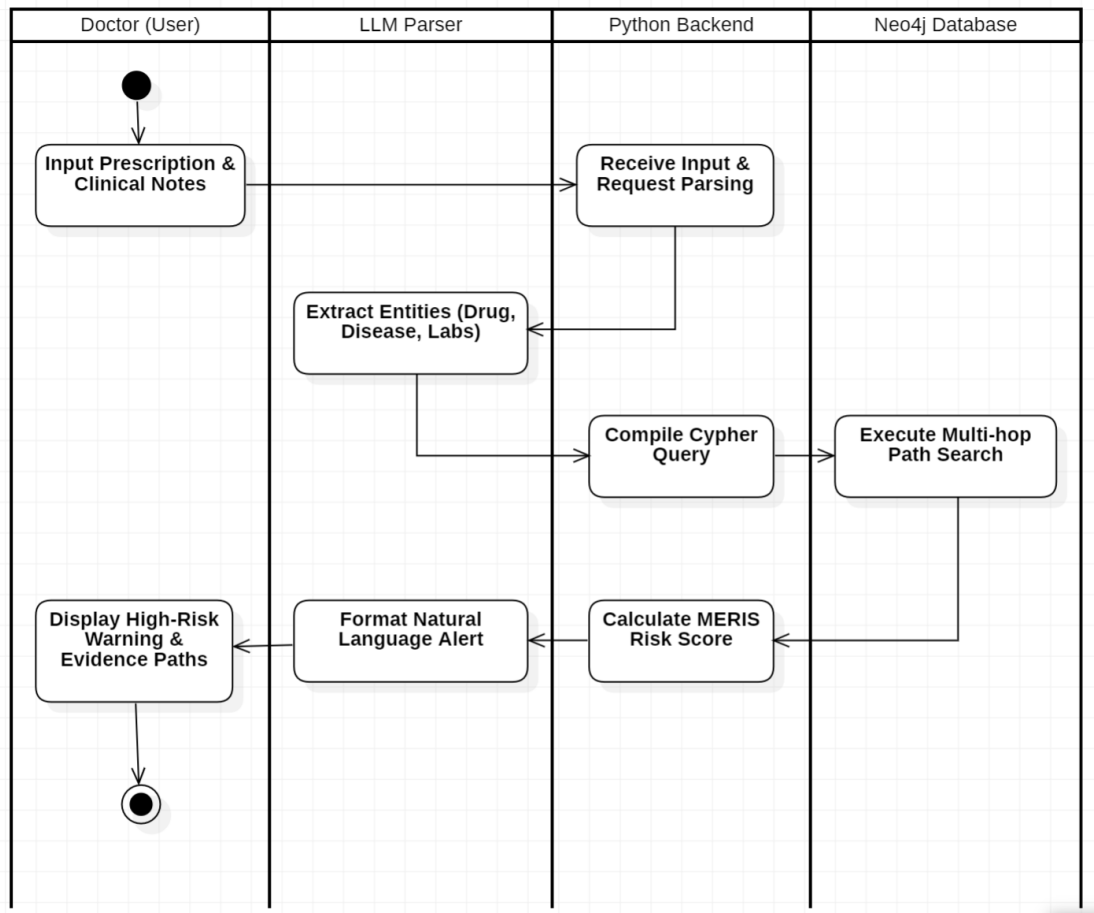


Figure 1 System Architecture and Data Workflow of the Proposed Graph-LLM CDSS

5.3 Testing and Evaluation

A set of prescription test cases is employed to measure the performance of the system. The following cases are initially run with the full graph and LLM setup. Then, the same inputs are sent to a regular LLM without any access to any graph. We can use the results of both runs to measure two things. The number of times the final warning text includes fabricated facts is tracked. Lastly, the computer's MERIS score is directly compared to correct answers which were hand calculated for each and every case.

5.4 Scope and Limitations

This prototype has 3 main boundaries. Synthea patient records and DrugBank rules are just fine for testing out the system design, but all of the data used in the project is fake. In reality, hospital records are much more complicated to deal with. Often it has no silly abbreviations or typing errors in synthetic files and real patients can have reactions to the body that aren't listed in medical books.

The search is restricted to three particular drug problems in heart patients within the graph. The first relates to drug and disease cross-reactions, for example, a person with known asthma receiving non-selective beta-blockers. The system also alerts on drug mixing problems. One excellent illustration of this is when a patient who is a user of Warfarin is also taking Aspirin. In the third focus, the conflicts with drugs or body tests are created such that the prescription becomes dangerous under the current body numbers. For example, ACE inhibitors are of concern in patients with recent lab results that reveal hyperkalemia.

On the software end, prescription notes are required to be a normal length and be structured. A long or illegible clinical story by a doctor may cause the LLM to be confused and decrease the accuracy of reading. All these restrictions mean that this project is an illustration of a software design concept rather than a system that is ready for use in a real hospital. We will demonstrate the system's way of thinking within those limits with a particular medical scenario, Risk of Hyperkalemia for Lisinopril. The system dances around (skips through) patient information, as shown in Figure 2, to identify this particular risk.

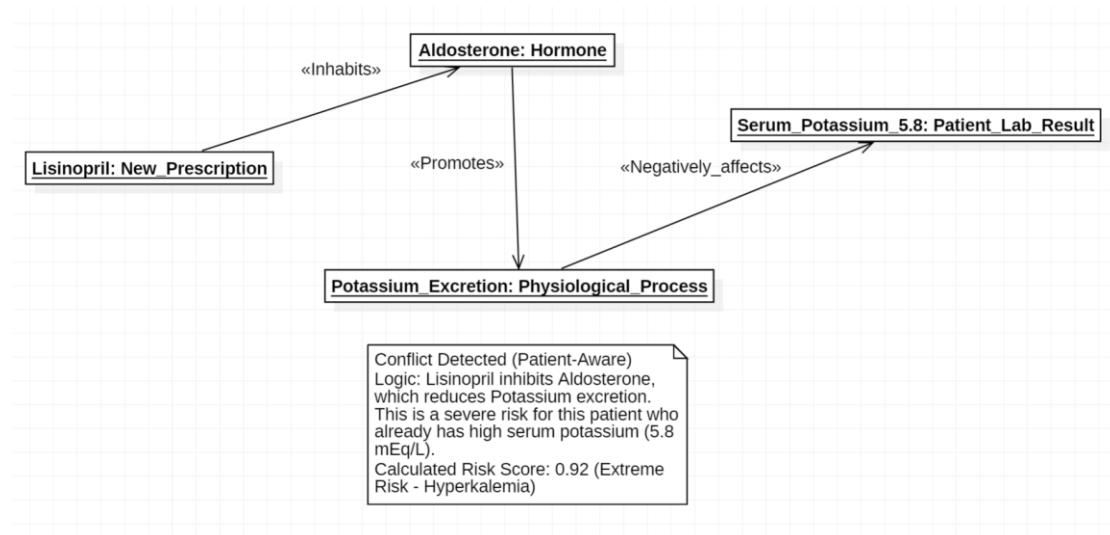


Figure 2 Object Diagram Illustrating Multi-hop Reasoning for Hyperkalemia Risk Detection

5.5 Ethical Considerations

Since all patient information is syned with Synthea, no real personal information is used in this project. This choice is known to have a limit. The true murky nature of real hospital cases is not reflected in fake records. This constrains our ability to use the results of the tests in everyday life. It's simply a proof of the design idea. It is not an instrument suitable for clinical use. All the results should be considered with this limitation in mind.

6. Potential Project Significance

Harm resulting from prescription errors — something hospitals can prevent. Existing tools for detecting such errors are not representative of the practice of medicine today. Patients information is recorded in regular human speech. The current CDSS programs can only read fixed tables. It's a very common issue that one can face. In hospitals it is normal to take several medicines together. Things get more complicated as patients age and receive more treatments. This project addresses this very gap

because with its LLM it reads text, but not makes medical decisions. The method is a safe process for normal typed clinical notes without the risk of fabricated facts being added to the prescription process. The knowledge graph then steps in, to perform the checking that flat tables cannot. It moves in and out of the various connected patient points in the graph to determine the actual patient's scenario. This general design is very clear in separating reading languages from the real facts. The system is designed to solve a larger problem with using AI in medicine since each job is assigned to the component with the best ability. A lot of new medical gadgets combine both these tasks and hence they fail the safety testing. Such separation creates a very secure system. This separation is so apparent that hospitals can track and explain how the software is making their clinical decisions.

7. Expected Outcomes and Concluding Remarks

The main result of this project is a working software prototype. It contains a Neo4j knowledge graph with only the drug connections and warning information provided by DrugBank. The prototype is also implemented with Python as the backend. This is a backend that generates synthetic patient data from Synthea. It interfaces with the LLM reading section and executes LLM graph searches as well as runs the MERIS math formula. The software has a web page for the clinic staff. They enter the prescription in there and are given simple warnings of risks.

The report will also be generated for the project that validates the functionality of the system in a particular test report. Inside this document, the accuracy of finding drug dangers is compared directly to real medical answers. We will also check if the computer MERIS score matches the human hand-calculated score. To measure the fake fact rates, the full Graph-LLM system runs against a normal LLM working alone on the exact same cases. These final numbers will show exactly how much the knowledge graph helps with accuracy. The data also points out where the LLM

reading part makes the most mistakes.

The system design itself stands as the other main result. Because the working software keeps a small open-source LLM strictly in a reading role, it gets all the real medical facts directly from a searchable knowledge graph. Other medical AI projects can easily copy this design idea. While this prototype is not ready for real hospital use, it successfully proves that joining these tools together actually works. Anyone can build the pipeline again using completely free tools. The project ultimately proves that this method is much safer than just using a language model alone to find prescription risks.

8. Key References

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9. Gantt Chart

This project requires two semesters to complete. The reading and planning are done by FYP1. This includes specifying the project scope, literature review and creating the system drawing. FYP2 is for actual coding. I will construct each system first, one at a time. Once that's done, I will tie them all together and perform the tests. I have left some extra time at the end of FYP2. I got these additional weeks to resolve any connection issues before the final submission and presentation. The complete Gantt Chart Schedule is presented in Figure 3 below. It provides you with all of the university deadlines and coding steps.

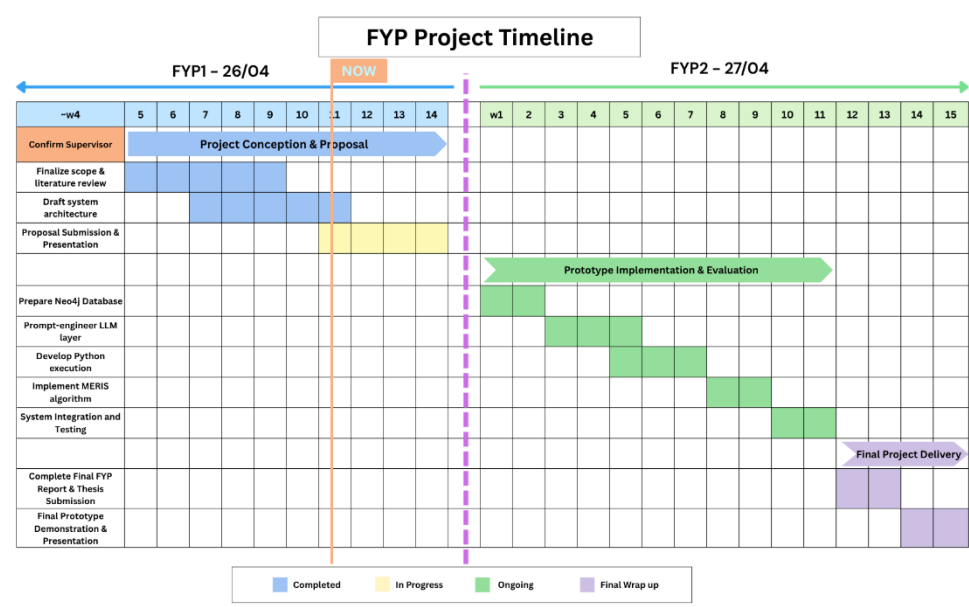


Figure 3 Gantt Chart for FYP

The project requires no special software or hardware. The coding and the local LLM are performed with a normal personal computer. The models in this system are actually very small but open-source and can be powered by a normal computer. I am using Neo4j Community Edition as my graph database. Back end code is written in Python and made use of free public libraries. The patient records are all from the free tool, Synthea, which are fake patient histories. These fake histories are virtually the

same as real medical records, except for the part with the patients. Drug warning data is from an open public drug database, called DrugBank. Only fake or public information is used by both data sources. They don't contain any actual patient information. This means the project does not need to apply for any data ethics approval. The project schedule will not have delays from any ethics checking process.

10. Appendix: Literature review synthesis matrix

Table 2. Literature review synthesis matrix

Theme & Citations	Research Objective	Core Concepts / Theories	Methodology & Tools	Evaluation Instrument	Key Findings & Contributions	Research Gap / Limitation	Application to Proposed Solution
Polypharmacy Risks & MERIS Scoring Lam et al. (2026) Saedder et al. (2015) Haji et al. (2021) Hertzian et al. (2025)	To validate medication risk algorithms (MERIS) and evaluate models for predicting harm in complex polypharmacy patients.	Clinical Risk Assessment; MERIS Algorithm; Polypharmacy	Retrospective cohort studies; Statistical modeling; Software CDSS development	AUC (Area Under Curve); Sensitivity; Automated calculation accuracy	Polypharmacy harm is highly prevalent. MERIS is a robust formula for risk quantification, and automating it via AI saves crucial clinical time.	Existing models struggle to scale safely with complex polypharmacy and often rely on static reasoning tasks.	Justifies the focus on polypharmacy risks. The Python module will implement the exact mathematical thresholds from the validated MERIS formula.

Theme & Citations	Research Objective	Core Concepts / Theories	Methodology & Tools	Evaluation Instrument	Key Findings & Contributions	Research Gap / Limitation	Application to Proposed Solution
LLM Strengths & Hallucination Risks Abumelha et al. (2025) Gapaia et al. (2024) Pal et al. (2023)	To evaluate LLMs' ability to parse clinical text and to benchmark their hallucination rates in medical inference.	LLM NLP processing; Causality extraction; AI Hallucinations	Two-phase LLM frameworks; Med-HALT benchmark testing	F1-scores; Precision/Recall; Hallucination scoring metrics	LLMs excel at parsing unstructured clinical notes but struggle with strict medical logic, frequently generating medically plausible but incorrect facts.	Focuses mostly on identifying hallucinations or extracting features, without proposing an architectural fix for decision-making.	Confirms that the LLM must be strictly restricted to a parsing role (translating notes to Python variables) and stripped of medical decision-making power.

Theme & Citations	Research Objective	Core Concepts / Theories	Methodology & Tools	Evaluation Instrument	Key Findings & Contributions	Research Gap / Limitation	Application to Proposed Solution
Knowledge Graphs for Clinical Reasoning Lin et al. (2023) Lyu et al. (2023) Zhou et al. (2022) Wishart et al. (2018)	To construct graph-based databases for multi-hop clinical reasoning, incorporating drug data and patient history.	Multi-hop reasoning; Patient-integrated graph modeling; Pharmacological structuring	KG construction; Relation extraction tools; Bioinformatics database (DrugBank) curation	Path-finding efficiency; Relationship mapping accuracy vs. relational DBs	Graph structures can encode clinical context (patient history) and successfully identify multi-hop, hidden contraindications missed by static tables.	Often focused on single disease domains and lack a user-friendly NLP interface for doctors' free-text inputs.	Neo4j will be populated with DrugBank data as the deterministic logic engine, dynamically receiving patient context to execute multi-hop Cypher queries.

Theme & Citations	Research Objective	Core Concepts / Theories	Methodology & Tools	Evaluation Instrument	Key Findings & Contributions	Research Gap / Limitation	Application to Proposed Solution
GraphRAG Framework Integration Lewis et al. (2021) Edge et al. (2025) Nygren et al. (2026)	To evaluate Retrieval-Augmented Generation (RAG) and GraphRAG architectures using lightweight LLMs for accurate retrieval.	Retrieval-Augmented Generation (RAG); Graph-guided retrieval; Compact Model Efficiency	Framework development; Comparative testing using compact LLMs (e.g., Qwen)	Exact Match (EM); F1 scores; Retrieval accuracy percentage	GraphRAG solves multi-entity connection problems. Lightweight, open-source models dramatically outperform standalone LLMs when paired with KGs.	Application is often general or focuses on static retrieval, lacking adaptation for dynamic, score-based patient systems.	Validates the overarching Graph-LLM architecture. Proves that a lightweight, open-source LLM is sufficient when paired with a traversable Knowledge Graph.

